Introduction to Vectors

Important Terminology

Most people have some idea of vectors since we encounter them daily. For example, you've likely seen your local meteorologist use vectors to describe wind speed and direction. In this lesson, we'll introduce some important terminology regarding vectors and approach them from a geometric perspective.

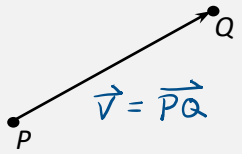
DEFINITION: ***Scalar and Vector Quantities, Resultants, Equilibrants, and the Zero Vector***

Quantities that can be described by ***magnitude*** alone, for example, speed, mass, temperature, etc., are called ___scalar___ quantities.

Quantities that have both ***magnitude*** and ***direction***, such as velocity, acceleration, etc., are called ___vector___ quantities.

Vector quantities are often represented by directed line segments called ___vectors___. The length of this vector denotes the magnitude of the vector quantity, while the orientation indicates the direction of the vector quantity.

A vector can be named by two points. The first point indicates the ___initial___ point or the ***tail***, and the second point indicates the ___terminal___ point or the ***head***.



The magnitude (or length) of any vector $\mathbf{v}$ is denoted as $|\vec{v}|$.

When two vectors have the same magnitude and direction, they are said to be ___equal___.

Vectors can be added together. The sum of any two vectors is called the ___resultant___, which is also a vector.

For any vector $\mathbf{v}$, a vector with equal magnitude but in the opposing direction is called an ___equilibrant___ and is denoted as $-\mathbf{v}$.

The sum of any vector $\mathbf{v}$, and its equilibrant $-\mathbf{v}$, is called the ___zero___ vector, denoted $\vec{0}$ (a vector quantity), and has a magnitude of 0 (a scalar quantity).

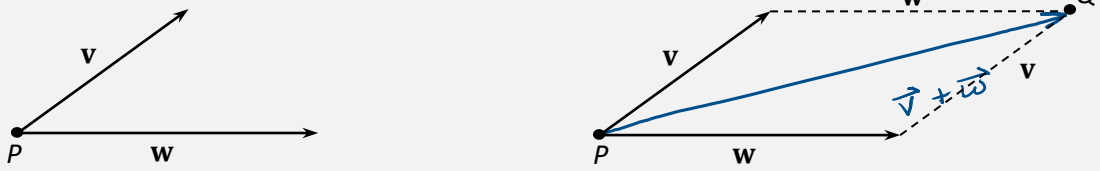
One common vector topic is that of force. In general, whenever we engage in an action that has the potential to accelerate an object, for example, pushing or pulling, we experience force.

Parallelograms and Vectors

Parallelograms are often used to visualize and understand vector operations, such as addition and subtraction.

IMPORTANT INFO: ***Parallelograms and the Resultant***

Given two vectors, $\mathbf{v}$ and $\mathbf{w}$, we can construct a parallelogram by arranging these vectors as opposite sides.



We visualize the resultant, $\mathbf{v} + \mathbf{w}$, as the directed diagonal formed with the initial side at the common initial point, point $\boldsymbol{P}$, and terminal side at the common terminal point, point $\boldsymbol{Q}$.

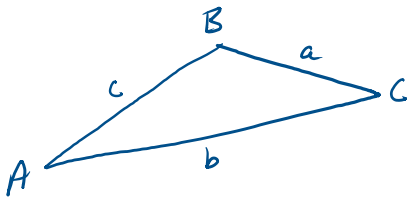
Note: *In a parallelogram, opposite sides are parallel and opposite angles are congruent. Additionally, we can conclude that adjacent angles in a parallelogram are supplementary.* (**Hint:** *From our parallel lines and transversals discussion, these angles are same-side interior.*)

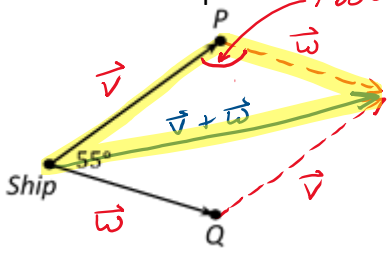
EXAMPLE 1

A ship is guided through a narrow river by two tugboats, as shown in the diagram below. The tugboat at point P can pull with a force of 60 thousand pounds, and the tugboat at point Q can pull with a force of 40 thousand pounds. Working together at full capacity, what is the magnitude of the resultant force, rounded to the nearest pound?

$180 - 55 = 125°$

$B = 125°$
$c = 60$
$a = 40$
$b = ?$





$b^2 = (40)^2 + (60)^2 - 2(40)(60)\cos(125°)$

$b^2 \approx 7953.166894$

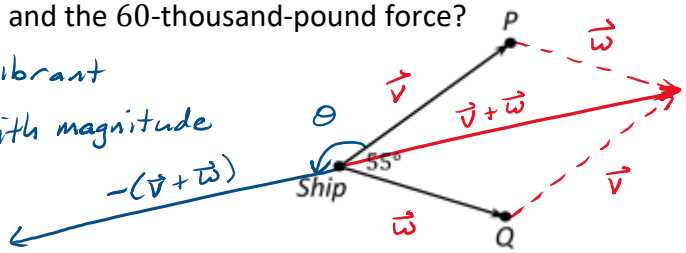
$b \approx 89.18052979 \longrightarrow |\vec{v} + \vec{w}| \approx 89{,}181\ lb$

EXAMPLE 2

Suppose the ship captain from Example 1 spots a small fishing boat in her path. She engages her engines in reverse to stop the forward motion and avoid a collision. What force must her engines apply to counter the pull of the tugboats? What angle is formed between this equilibrant force and the 60-thousand-pound force?

Must apply an equilibrant force $\vec{F} = -(\vec{v} + \vec{w})$ with magnitude

$|\vec{F}| = |-(\vec{v} + \vec{w})| \approx 89.181$ lb

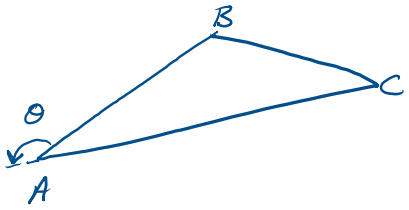


$\frac{\sin A}{a} = \frac{\sin B}{b}$

$\sin A \approx \frac{40 \sin(125)}{89.181} \approx 0.367411015$

$A \approx 21.56° \rightarrow \theta \approx 180 - 21.56$

$\theta \approx 158.44°$



$B = 125$

$c = 60$

$a = 40$

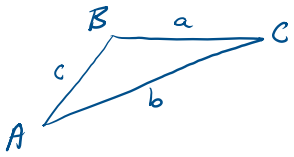
$b \approx 89.181$

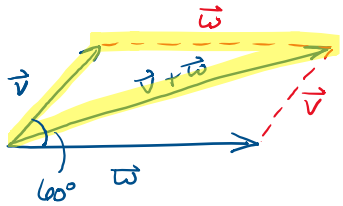
$A \approx ?$

EXAMPLE 3

Consider two vectors, $\mathbf{v}$ and $\mathbf{w}$, which share a common initial point and form an angle of 70°. If $|\mathbf{v}| = 6$ in, and $|\mathbf{v} + \mathbf{w}| = 11$ in, sketch $\mathbf{v}$, $\mathbf{w}$, the resultant $\mathbf{v} + \mathbf{w}$ using a parallelogram, and find $|\mathbf{w}|$.

We know	We need
$B = 110$	$C =$
$c = 6$	$A =$
$b = 11$	$a =$





$\sin C = \frac{c \sin B}{b} = \sin^{-1}\left(\frac{6 \sin(110)}{11}\right) \approx 30.83°$

$A \approx 180 - 110 - 30.83 \approx 39.17$

$a = \sin A \left(\frac{b}{\sin B}\right)$

$a \approx \sin(39.17)\left(\frac{11}{\sin(110)}\right) \approx 7.39$

$|\vec{w}| \approx 7.39$ in

Inclined Planes and Right Triangles

When an object is positioned on an inclined plane, the force of gravity acts downward, thus moving the object in the downhill direction. Therefore, to remain at rest, another force must be applied to counter the force of gravity (for this text, we will ignore any effects of friction). In examples like this, we often construct a diagram and find opportunities to utilize right triangle trigonometry.

EXAMPLE 4

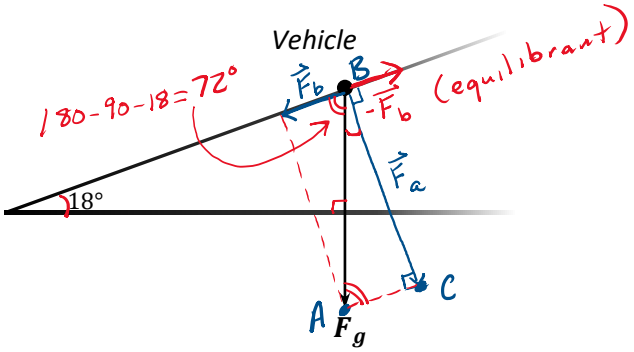
A 3800-lb vehicle is held steady on a ramp that makes an 18° angle with the horizontal by a rope tied to a secure block further uphill. A diagram (not to scale) is provided to help visualize the angles and forces involved.

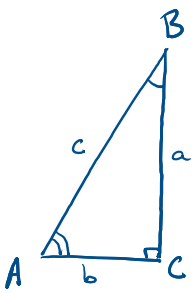
(a) Construct a parallelogram with $\mathbf{F_g}$ (force due to gravity) as a diagonal by adding two vectors to this diagram.

(i) Sketch a vector, $\mathbf{F_b}$, parallel to the ramp surface oriented in the downhill direction.
(ii) Sketch another vector, $\mathbf{F_a}$, perpendicular to the ramp surface in a general downward direction.

These two vectors illustrate the decomposition of $\mathbf{F_g}$ into two components. In our next lesson, we will explore vector components in greater detail.

(b) Label the head and tail of the vector $\mathbf{F_g}$ as points A and B, respectively. Label the head of $\mathbf{F_a}$ as point C. You should now have the triangle, ABC. Identify and justify each angle measure.





$B = 18°$
$A = 72°$
$C = 90°$
$b = |\vec{F}_b|$
$a = |\vec{F}_a|$
$c = |\vec{F}_g| = 3800$

(c) From our parallelogram, we recognize that $\mathbf{F_a} + \mathbf{F_b} = \mathbf{F_g}$. We can also see that the vector, $\mathbf{F_b}$, is the force we must overcome to keep the vehicle at rest. That is, we must find the equilibrant, $-\mathbf{F_b}$. Sketch this equilibrant in the diagram above.

(d) Assume the rope used has a maximum safe working load rating of 1300 lb. Using your results from Part (b), can this rope be trusted to hold the car at rest?

$\sin(B) = \frac{b}{c} \rightarrow b = c\sin(B) = 3800\sin(18°) \approx 1174.3$

Yes, $|\vec{F}_b| \approx 1174.3$ lb < 1300 lb (safe working load)

(e) What is the maximum ramp angle at which this rope can safely hold this vehicle in place?

$\sin(B) = \frac{b}{c} \rightarrow B = \sin^{-1}\left(\frac{b}{c}\right) = \sin^{-1}\left(\frac{1300}{3800}\right)$ ← Max safe working load

$B \approx 20.0°$

Max ramp angle